

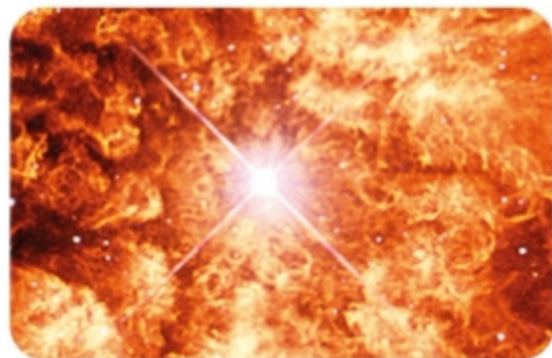
H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og
		La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Im	Yb	Lu	
		Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

The noble gases sit in group 18—the final group of the periodic table.

This sample of helium is stored in a glass sphere.



Helium is a colorless gas, but it glows purple when electrified.



Wolf-Rayet 124 glows inside the M1-67 Nebula.

Noble Gases

Because they are colorless, odorless, tasteless, and mostly unreactive, the noble gases were not easy to identify. But once one was found, the rest followed fairly quickly. Argon was the first of the family to be discovered by the British chemists Morris William Travers and William Ramsay, in 1894. With the subsequent discoveries of helium, krypton, neon, and xenon, a whole new group on the periodic table was established.

Helium

²He Although abundant in the universe, helium is not widespread on Earth. The sun and other stars contain a lot of helium, produced by pairs of hydrogen atoms crashing into one another to make helium atoms while releasing massive amounts of energy. On Earth, helium is produced naturally, but very slowly, as a result of radioactive decay.

Keeping cool

In a particle accelerator (a machine that smashes atoms together in scientific experiments), liquid helium is injected into the electromagnets to keep the magnets cool. The Large Hadron Collider (LHC) at CERN—the European Organization for Nuclear Research on the French-Swiss border—is the world's biggest particle accelerator.



Extreme-helium stars

Unlike most stars that are composed primarily of hydrogen, extreme-helium stars such as Wolf-Rayet 124 (left) are made up of approximately 85 percent helium. Wolf-Rayet 124 is an ultra-massive star surrounded by the intensely hot M1-67 Nebula. This type of star also contains smaller amounts of other elements, including carbon, nitrogen, oxygen, aluminum, iron, chromium, and nickel.